

Introduction to SPICE simulator

What is this SPICE all about?

- Stands for Simulation Program with Integrated Circuit Emphasis
- Used for simulating circuits for verifying performance and optimization
- Powerful simulator with industry standards
- Supports different types of analysis like AC, DC, TRAN, PZ, OP etc.

Why SPICE

- Circuits get complex with increasing number of transistors
- Analytically solving is impractical
- Even for small circuits in scaled technologies the number of parameters are far too many
- Example
 - The BSIM4 MOS model has about 39 technology dependent parameters!!

SPICE Code Format

- The SPICE code includes following sections:

TITLE LINE

CIRCUIT DESCRIPTIONS

.MODEL STATEMENTS

ANALYSIS COMMANDS

OUTPUT COMMANDS

.END

- The **title line** and is ignored by the simulator. This line is printed verbatim on the output screen
- Comment lines should be preceeded with an *.
- Multiple files can be included using **.include** directive. These other files may be subcircuits or model files for custom devices.

SPICE Code Format

- The “**analysis commands**” include indications for transient, DC , AC analysis.
- the “**output command**” must begin with **.control** directive and terminates with **.endc** directives.
- Commands not written in the **.control** block should be preceded with a “.”
- The “**output commands**” are used for saving or plotting or printing outputs.
- The program file should end with a **.end** directive.

Circuit Description

- The **CIRCUIT DESCRIPTION** in SPICE is called “netlist”, consisting of element and source description.
- Element description format:
`<letter><name> <n1> <n2>... [modelname] [parvals]`
- Elementary components in native spice are
 - R^* => resistor
 - C^* => capacitor
 - L^* => inductor
 - SW^* => switch
 - K^* => coupled mutual inductor
 - M^* => MOSFET
nodes: [DRAIN -> GATE -> SOURCE-> SUBSTRATE]
 - Q^* => BJT
nodes: [COLLECTOR->BASE->EMITTER->SUBSTRATE]

- Elementary sources :

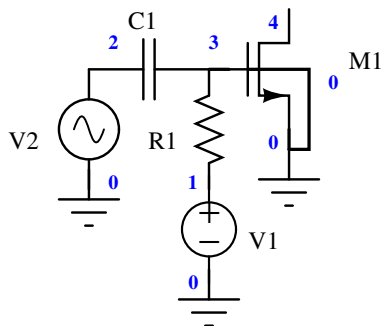
Independent sources

- $V\langle\text{name}\rangle \langle\text{n+}\rangle \langle\text{n-}\rangle [\text{type}] \langle\text{val}\rangle$
- $I\langle\text{name}\rangle \langle\text{n+}\rangle \langle\text{n-}\rangle [\text{type}] \langle\text{val}\rangle$

Dependent sources

- $\langle\text{letter}\rangle\langle\text{name}\rangle \langle\text{nout+}\rangle \langle\text{nout-}\rangle \langle\text{nc+}\rangle \langle\text{nc-}\rangle \langle\text{gain}\rangle$
 - $G^* \Rightarrow$ linear voltage controlled current source
 - $E^* \Rightarrow$ linear voltage controlled voltage source
 - $F^* \Rightarrow$ linear current controlled current source
 - $H^* \Rightarrow$ linear current controlled voltage source
- Nonlinear dependent sources are also available (look at references if interested)

Circuit Description: Example



R1 3 1 1K

C1 2 3 1p

M1 4 3 0 0 cmosn W=2um L=0.18um

V1 1 0 dc 5V

V2 2 0 sin(0 1V 1K 0 0)

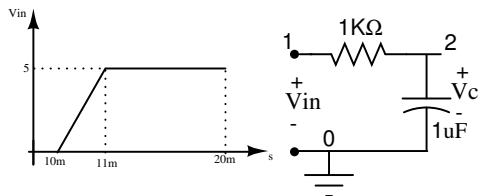
Analysis Commands

- DC analysis for DC transfer characteristics
 - .DC source_name vstart vstop stepsize
- AC analysis for frequency response
 - .AC DEC/OCT/Lin NP fstart fstop
 - .AC DEC 10 10 100M
- Transient analysis for time response
 - .TRAN stepsize stoptime starttime

Output Commands

- Plot voltages and currents
 - `plot V(1) V(2) I(3) Vid#branch`
 - `plot V(3) vs V(4)`
- Save vectors in file
 - `set hcopydevtype=postscript`
`hardcopy filename.ps V(1) V(2) ..`
- Print voltages and currents
 - `print V(vout) V(1)`

Example:RC Circuit transient analysis



RC Transient Response

*resistor connected between nodes 1 and 2

r1 1 2 1k

*capacitor connected between nodes 2 and 0

c1 2 0 1u

*piecewise linear input voltage

vin 1 0 pwl (0 0 10ms 0 11ms 5v 20ms 5v)

Example:RC Circuit transient analysis

```
*transient analysis for 20ms, step size 0.02ms
.tran 0.02ms 20ms
*defining the run-time control functions
.control
run
*plotting input and output voltages
plot v(1) v(2) i(v1)
.endc
.end
```

Example:RC Circuit AC analysis

For the same R-C circuit let us do the small-signal AC analysis.

RC Circuit Frequency Response

```
r1 1 2 1k
```

```
c1 2 0 1u
```

*Specifying an AC source with zero dc

```
vin 1 0 dc 0 ac 1
```

*AC analysis for 1 Hz to 1MHz, 10 points per decade

```
.ac dec 10 1 1Meg
```

```
.control
```

```
run
```

*Magnitude dB plot for v(2) on log scale

```
plot vdb(2)
```

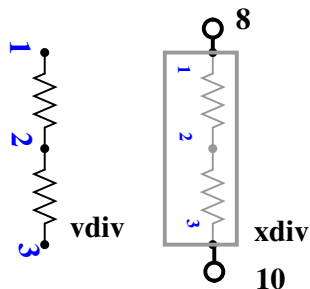
*Phase degrees plot for v(2) in degree on log scale

```
let phase=180/PI*vp(8) plot phase
```

```
.endc
```

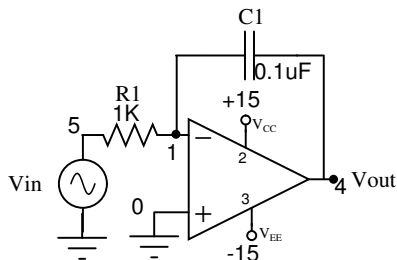
```
.end
```

Subcircuits



- Regular circuits can be described using repeated instantiations of elemental cells
 - Subcircuits description
 - .subckt nodes sname
 - ...netlist
 - .ends sname
 - Subcircuits are identified by X* port list instance name
 - X* nodes sname
- ```
xdiv 8 10 vdiv
```

# Example: Integrator using OPAMP-741



Integartor using op-amp 741

\*Including the predefined op-amp subcircuit file

.include ua741.txt

\*Connections as mentioned in subcircuit file

X1 0 1 2 3 4 UA741

R1 5 1 1k

C1 4 1 0.1u

# Example: Integrator using OPAMP-741

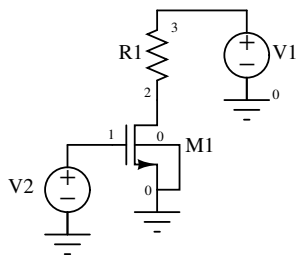
```
vcc 2 0 dc 15v
vee 3 0 dc -15v
*Giving a sinusoidal input
vin 5 0 sin (0 1v 1k 0 0)
.tran 0.02ms 6ms
.control
run
plot v(5) v(4)
.endc
.end
```



# .MODEL STATEMENTS

- Devices in a circuit are defined by same set of device model parameters.
- These parameters are defined on a separate **.model** line and assigned a unique model name.  
`model mname type(pname1=pval1 pname2=pval2 . .)`
- Model parameters written in a file can also be included using **.include** command.

# NMOS Amplifier



MOS amplifier- with .MODEL card

```
***** NETLIST*****
```

```
M1 2 1 0 0 MOSN W=2um L=5um
```

```
.model MOSN NMOS level=8 version=3.3.0
```

```
R1 3 2 1K
```

```
V1 3 0 1.8V
```

```
V2 1 0 0.9V
```

```
.END
```

MOS amplifier-with model file

```
.include tsmcmodel.txt
```

```
***** NETLIST*****
```

```
M1 2 1 0 0 cmosn W=2um L=5um
```

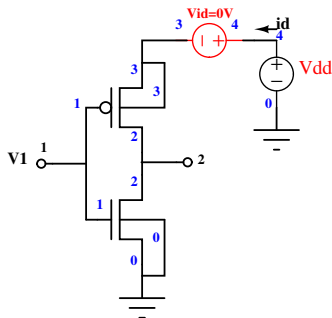
```
R1 3 2 1K
```

```
V1 3 0 1.8V
```

```
V2 1 0 0.9V
```

```
.END
```

# Example: CMOS Inverter



## DC analysis

- \* include model file

```
.include tsmc_spice_model.txt
```

- \*Element instantiation

```
M1 2 1 0 0 cmosn w=2um l=0.18um
```

```
M2 2 1 3 3 cmosp w=5um l=0.18um
```

- \*supply voltages , input sources

```
vdd 4 0 dc 1.8V
```

```
vin 1 0 0.0 pulse (0 1.8v 0 0 0 5us 10us)
```

- \*dummy voltage to measure current

```
vid 4 3 dc 0v
```

# Example: CMOS Inverter

```
*perform DC analysis
.dc Vin 0 1.8 0.1
.control
run
plot v(2)
plot v(1)
plot i(vid)
plot v(2) vs v(1)
*save the plots in eps file
set hcopydevtype=postscript
hardcopy invplots.eps i(vid)
.endc
.end
```

# Example: CMOS Nand

Nand transient analysis

\* include model file

.include tsmc\_spice\_model.txt

\*Element instantiation

m1 3 2 0 0 cmosn w=2um l=0.18um

m2 4 1 3 0 cmosn w=2um l=0.18um

m3 4 1 5 5 cmosp w=5um l=0.18um

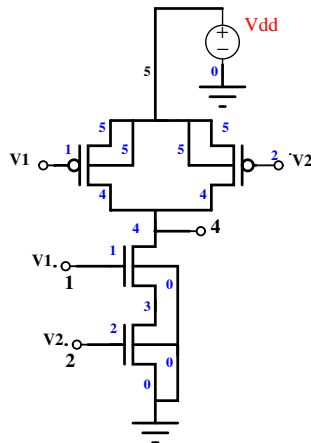
m4 4 2 5 5 cmosp w=5um l=0.18um

\*supply voltages,input sources

vdd 5 0 dc 1.8v

v1 1 0 pulse (0 1.8v 0 1ns 1ns 5us 10us)

v2 2 0 pulse (0 1.8v 0 1ns 1ns 2.5us 5us)



# Example: CMOS Nand

```
*perform transient analysis
.tran 0.02us 10us
.control
run
plot v(4)
plot v(1) v(2)
.endc
.end
```

# Example: CMOS AND gate using subcircuits

AND transient analysis

\* NAND subcircuit

```
.subckt inv 1 2 5
```

```
.include tsmc_spice_model.txt
```

```
m1 2 1 0 0 cmosn w=2um l=0.18u
```

```
m2 2 1 5 5 cmosp w=2um l=0.18u
```

```
.ends inv
```

\* NAND subcircuit

```
.subckt nand 1 2 4 5
```

```
.include tsmc_spice_model.txt
```

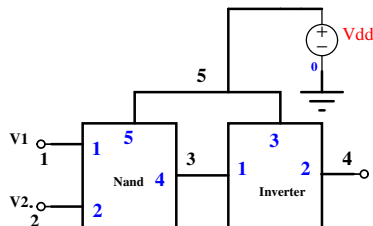
```
m1 3 2 0 0 cmosn w=2um l=0.18u
```

```
m2 4 1 3 0 cmosn w=2um l=0.18u
```

```
m3 4 1 5 5 cmosp w=2um l=0.18u
```

```
m4 4 2 5 5 cmosp w=2um l=0.18u
```

```
.ends nand
```



# Example: CMOS AND gate using subcircuits

```
*instantiate the subcircuits
xnand 1 2 3 5 nand
xinv 3 4 5 inv
vdd 5 0 dc 1.8v
V1 1 0 pulse (0 1.8v 0 1ps 1ps 5us 10us)
V2 2 0 pulse (0 1.8v 0 1ps 1ps 2.5us 5us)
*specify analysis
.tran 0.02us 10us
.control
run
plot v(4)
plot v(1) v(2)
.endc
.end
```